

Diagnostic Evaluation Report

1. Framework Information

- **Framework:** X-RAG Diagnostic Framework (v1.0)
- **Trace ID:** `d72f8394-be9c-4f9f-a8d8-55978a9b5d84`
- **Timestamp:** 2026-09-05T16:05:44.146238Z
- **Status:** COMPLETED

2. Executive Summary

- **Question:** Which judgments in this corpus apply or construe section 140?
- **Answer:** I do not have enough information to answer this. The retrieved chunks (16) contain only statutory provision 140 of the Bharatiya Nyaya Sanhita, 2023.
- **Health Score:** 0.76
- **Primary Issue:** UNKNOWN
- **Claims Supported:** 2 / 3
- **Summary:** Analysis completed.

3. Pipeline Overview

Stage	Status	Confidence	Observation
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CORPUS	PASS	0.9	Corpus contains sufficient relevant content, evidenced by at least one supported claim.
RETRIEVER	PASS	0.95	At least one retrieved chunk provides supporting evidence for a verified claim.
GENERATOR	UNKNOWN	0.5	1 generated claims are unsupported, but retrieval quality is ambiguous.
GROUNDING	PASS	0.9	No claims are contradicted by retrieved evidence.

Efficiency: 2/6 chunks used (33%)

4. Evaluation Metrics

- **Retrieved Chunks:** 6
- **Verified Claims:** 3 (Supported: 2, Partial: 0, Unsupported: 1, Contradicted: 0)
- **Grounding Score:** 0.67
- **Evidence Coverage:** 0.33
- **Average Entailment:** 0.58
- **Latencies:** Retrieval (2374.3ms) | Generation (17194.7ms) | Verification (10007.8ms)

5. RAGAS Metrics

- **Faithfulness:** 0.6667
- **Answer Relevancy:** 0.7372
- **Context Precision:** 1.0000
- **Context Relevancy:** 0.1667
- **Context Recall:** N/A
- **Answer Similarity:** N/A
- **Answer Correctness:** N/A

Answer Correctness (Claim Recall)

Not computed for this trace.

6. Evidence Analysis

- **[1bde8ae3-7eb1-492b-9084-c237ad78e06a]** (UNSUPPORTED): I do not have enough information to answer this.
- *Evidence (Chunk eb20b0ca-98af-4633-b67e-49a6e31e8f9e):* 1]

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- **[b797fc65-350c-4724-a56d-e3b6e4e83fb0]** (SUPPORTED): The retrieved chunks (16) contain only statutory provisions and
- *Evidence (Chunk cbfadd60-b8b0-45f1-8f66-7f75b4f36aa2):* Any person,—
(i) who has been ordered under section 136 to give security for keeping the

peace or for good behaviour; or
(ii) who is aggrieved by any order refusing to accept or rejecting a surety under section 140,
may appeal against such order to the Court of Session:
Provided that nothing in this section shall apply to persons the proceedings against whom are laid before a Sessions Judge in accordance with the provisions of sub-section (2) or sub-section (4) of section 141.

- **[a45760aa-cfd5-44cc-842e-0f2c1a5330e1]** (SUPPORTED): None of the retrieved chunks include any judgments that apply or 140 of the Bharatiya Nyaya Sanhita, 2023.

- **Evidence (Chunk eb20b0ca-98af-4633-b67e-49a6e31e8f9e):** 1]

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7. Root Cause Analysis

- **Primary Cause:** UNKNOWN
- **Confidence:** 0.92
- **Reasoning:**
 - Starting root cause analysis for trace d72f8394-be9c-4f9f-a8d8-55978a9b5d84.
 - CORPUS passed (confidence=0.90).
 - RETRIEVER passed (confidence=0.95).
 - Skipping CHUNKING: no observable state in matrix.
 - Skipping GENERATOR: status is UNKNOWN (insufficient evidence).
 - GROUNDING passed (confidence=0.90).
 - No failures detected. Pipeline is healthy or all evidence is UNKNOWN.

8. Corrective Action Plan

- **[INFORMATIONAL] Review Retrieval Top-K / Reranker Cutoff**
 - Description: Retrieval efficiency is low (33%); consider reducing top-k or tightening the reranker score cutoff.
 - Expected Improvement: Reduced prompt size and lower latency/cost with no loss of verified support. (Metric: Chunk utilization)

9. Evidence Traceability

- **Primary Cause:** UNKNOWN
- **Derived From:** RootCauseAnalysis
- **Pipeline Stage:** UNKNOWN
- **Supporting Claims:** 1bde8ae3-7eb1-492b-9084-c237ad78e06a, b797fc65-350c-4724-a56d-e3b6e4e83fb0, a45760aa-cfd5-44cc-842e-0f2c1a5330e1
- **Supporting Chunks:** eb20b0ca-98af-4633-b67e-49a6e31e8f9e, cbfadd60-b8b0-45f1-8f66-7f75b4f36aa2, eb20b0ca-98af-4633-b67e-49a6e31e8f9e

10. Overall Assessment

- **Major Strength:** RETRIEVER is healthy (confidence=95%): At least one retrieved chunk provides supporting evidence for the claim.
- **Major Weakness:** UNKNOWN
- **Next Priority:** Review Retrieval Top-K / Reranker Cutoff
- **Recommendation:** Pipeline requires targeted improvements to the primary failure stage.

11. Framework Footer

- **Generated By:** X-RAG Diagnostic Framework
- **Artifact Version:** 1.0

APPENDIX

Artifacts Generated

✓ RAGTrace
artifacts/rag_traces/d72f8394-be9c-4f9f-a8d8-55978a9b5d84.json

✓ ClaimSet

artifacts/claim_sets/d72f8394-be9c-4f9f-a8d8-55978a9b5d84.json

✓ VerificationResults

artifacts/verification/d72f8394-be9c-4f9f-a8d8-55978a9b5d84.json

✓ PipelineStateMatrix

artifacts/pipeline_state_matrix/d72f8394-be9c-4f9f-a8d8-55978a9b5d84.json

✓ RootCauseAnalysis

artifacts/root_cause_analysis/d72f8394-be9c-4f9f-a8d8-55978a9b5d84.json

✓ CorrectiveActionPlan

artifacts/corrective_action_plan/d72f8394-be9c-4f9f-a8d8-55978a9b5d84.json

✓ DiagnosticEvaluationReport

artifacts/reports/d72f8394-be9c-4f9f-a8d8-55978a9b5d84.json